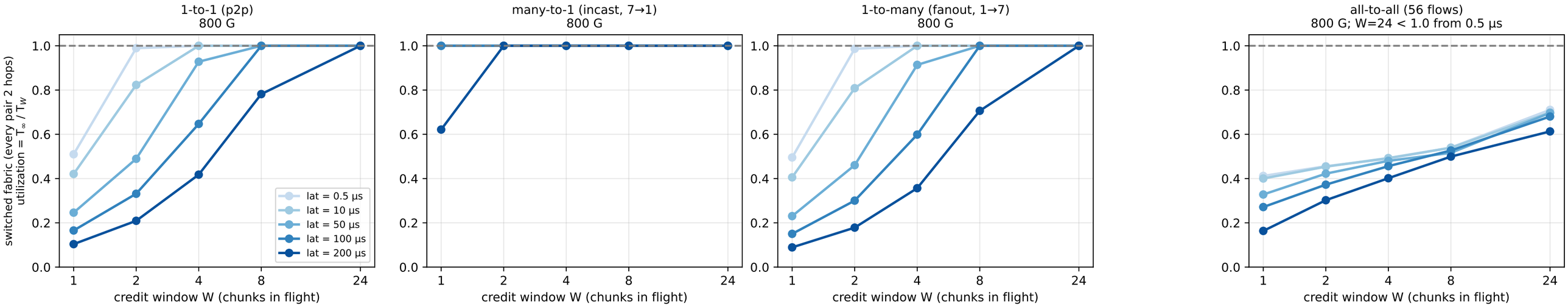
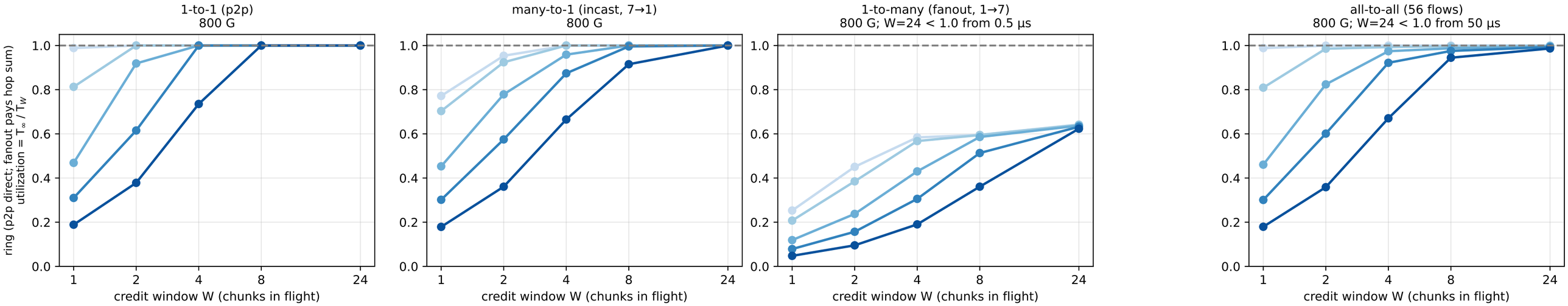
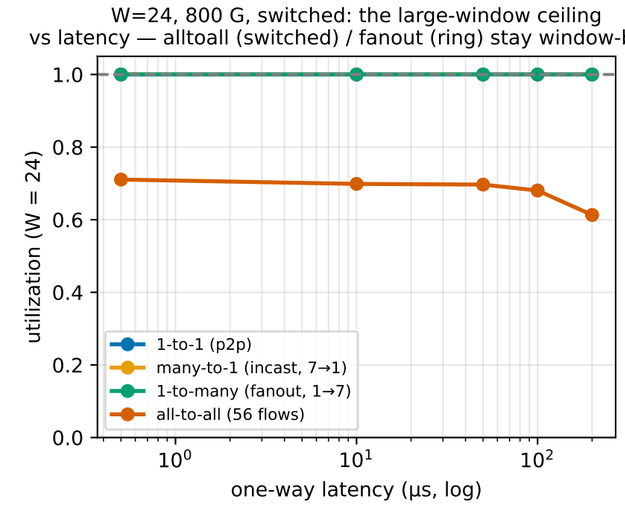
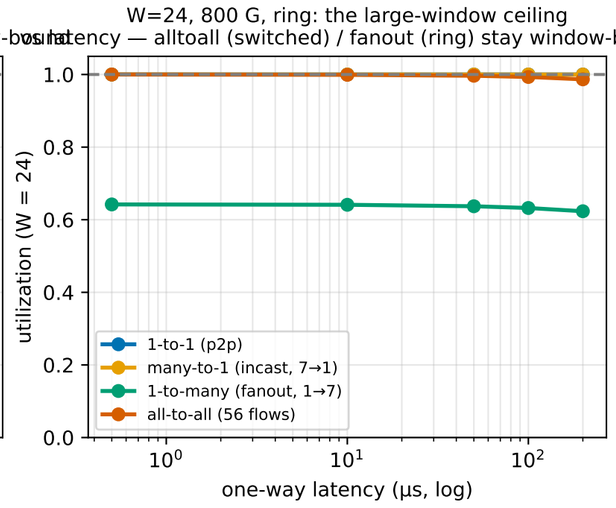
Credit flow control across traffic patterns at FIXED bandwidth (800 G, 8-GPU Su fabric, 256 MiB per flow):
utilization vs credit window W and one-way latency — dedicated (p2p/incast) vs shared (fanout/alltoall) windows, ring vs switched









Utilization at 800 G, W = 24

lat (µs):	0.5	10	50	100	200
--- switched ---					
1-to-1 (p2p)	1.000	1.000	1.000	1.000	1.000
many-to-1 (incast, 7→1)	1.000	1.000	1.000	1.000	1.000
1-to-many (fanout, 1→7)	1.000	1.000	1.000	1.000	1.000
all-to-all (56 flows)	0.711	0.699	0.697	0.680	0.613
--- ring ---					
1-to-1 (p2p)	1.000	1.000	1.000	1.000	1.000
many-to-1 (incast, 7→1)	1.000	1.000	1.000	1.000	1.000
1-to-many (fanout, 1→7)	0.642	0.641	0.637	0.632	0.623
all-to-all (56 flows)	1.000	0.999	0.997	0.993	0.987

Reading rows 1-2: the W needed for full utilization shifts right ~linearly with latency (credit RTT = hops·(ser+2·lat)); fanout/alltoall need ~W×flow-count credits (shared 1-VC window), so even W=24 stays window-bound at 200 µs — the shared window, not the link, is the bottleneck.

